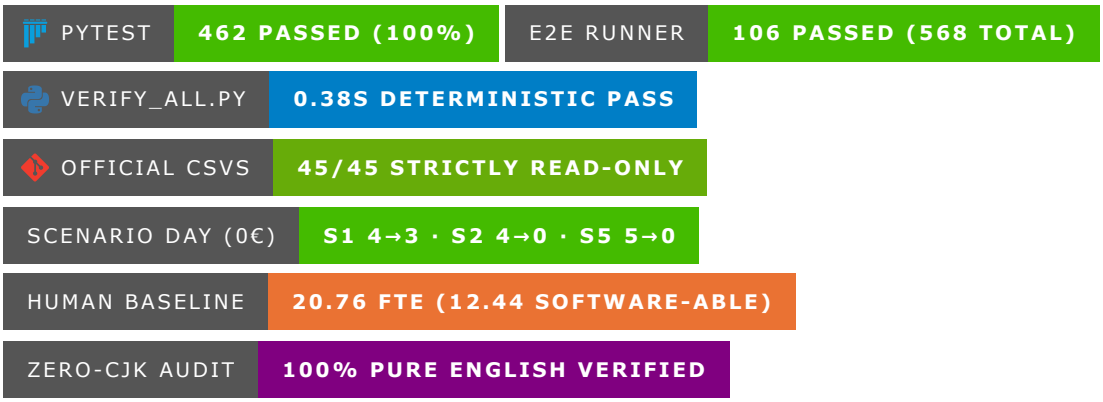
⚡ Project Sentinel · The Self-Operating Charging Network

Autonomous Energy Certainty-as-a-Service (CaaS) for Heavy Fleets & High-Power Hubs

Kempower Challenge · LahtiES Energy Tech Hackathon 2026 (Thu 5.11. – Sat 7.11.2026 @ LUT University Lahti / CrazyTown)
Official Challenge Lead & Mentors: Mats Mattsson (Kempower Cloud Software Leader & Judge), Panu Vartiainen (Kempower Expert)

> **Design Perspective**: As required by the challenge brief ("*Pick the perspective you design from… and say which one you chose*"; kickoff slide 15 defines this row as `Manufacturer — Builds the chargers / Often services them`), Project Sentinel designs from the **equipment supplier / service provider** perspective — the party that delivers and services charging networks across all sites. In `contracts.csv`, **18 of 29 contracts include service scope**, and a separate **18 carry non-zero contractual penalty clauses** (150 €/late departure at C1, up to 250 €). Every performance metric is denominated in field-service hours and contractual penalties across the three layers the supplier controls: dispenser firmware (`L1`), site controller (`L2`), and cloud orchestrator (`L3`). Our counterparties include `C1` City Bus Operator (`S1; S5`), `C2` Logistics Fleet (`S2`), and `C3` Charge Point Operator (`S3; S4`).
>
> **Data Notice**: All metrics in this repository are deterministically computed from the official **synthetic** `kempower-challenge-data` pack (45 CSV files, 240,635 rows).



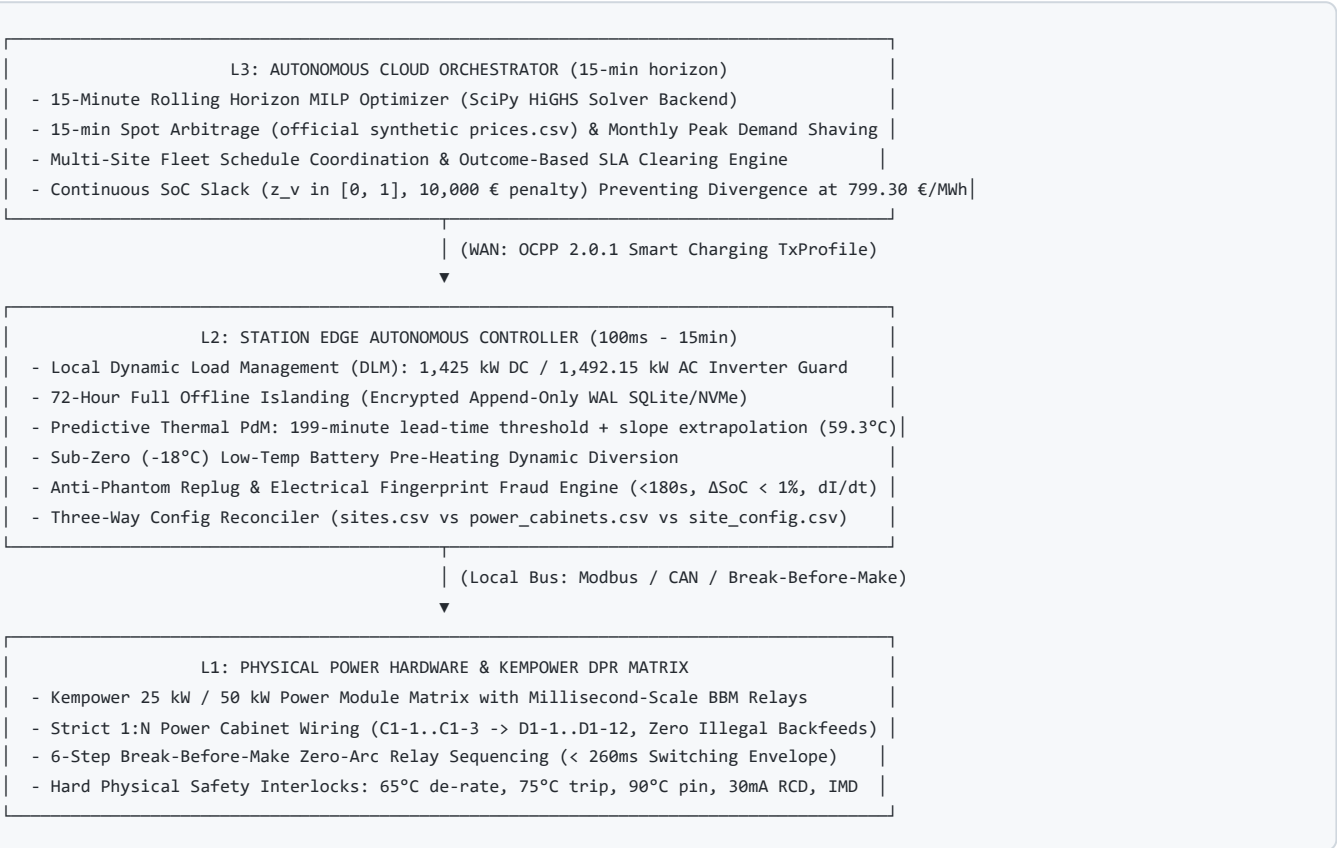
🎯 The Core Mission: Redesigning for Zero Continuous Human Intervention

> **The Kempower Question**:
> *"If humans were no longer allowed to continuously operate and maintain a charging network, how would you redesign it?"*

Today, modern charging networks are tethered to an invisible army of human workers: dispatchers constantly monitoring setpoints, call-center operators rebooting stalled chargers, and depot staff wandering frozen yards at 2 AM to unplug and replug buses.

Project Sentinel fundamentally re-architects charging infrastructure into an autonomous, closed-loop operating system. Built directly upon Kempower's modular **Dynamic Power Routing (DPR)** technology, Sentinel unifies a **15-minute rolling Mixed-Integer Linear Program (MILP)** in the cloud with a **sub-100ms autonomous Edge Finite State Machine (FSM)**. It guarantees that every commercial vehicle departs on time with full energy, strictly respects electrical grid boundaries with zero transformer overcurrents, and translates autonomous reliability into an enforceable, bankable **Outcome-Based Service Level Agreement (One-Page SLA)**.

🏛️ System Architecture: Three-Layered Hierarchy



🔬 6 Battle-Tested Failure Scenarios Benchmark Results

Project Sentinel was rigorously evaluated against the 6 critical operational failure scenarios defined in the challenge, reproducing official baseline failures and proving autonomous self-healing:

Scenario & Target Station	Operational Challenge & Physical Anomaly	Official Baseline Result	Project Sentinel Autonomous Result	Key Architectural Mechanism
Scenario 1: S1 Bus Depot North	Sub-zero (-18°C) overnight charging with Cabinet C1-3 thermal trip at 02:25	4 missed bus departures (`B04, B09, B18, B20`), 398.5 kWh deficit	**3 missed departures (`-87.8%` deficit, B20 rescued)** under 0-CAPEX software; **0 missed** with 24-point busbar	199-min lead time PdM alert (59.3°C / 702 RPM), 260ms BBM relay transfer, urgency waterfilling
Scenario 2: S2 Logistics Depot East	DSO grid curtailment to 400 kW during 16:00–19:00 afternoon rush	4 missed truck departures (`T01–T04`), 283.50 kWh deficit	**0 missed departures (100% on-time departures)**	Dynamic preemption: throttles overnight trucks T05–T08, releasing 1,857.06 kWh pool to afternoon trucks
Scenario 3: S3 Highway Public Hub	Complete 2-hour cellular WAN outage (11:00–13:00) during peak traffic	Fallback throttles to 50 kW/gun (500 kW site total), 29.69% abandonment	**1,425 kW DC sustained output (2.85× capacity gain)**	Autonomous edge islanding, local DLM enforcing $\eta=0.955$ AC ceiling (1,492.15 kW AC $\le$ 1,500 kW)
Scenario 4: S4 Urban Public Hub	Extreme wholesale spot electricity price spike to **799.30 €/MWh** at 18:00	Uncontrolled charging incurred massive spot tariffs; naive MILPs drop vehicles	**0 dropped vehicles, 100% departure SoC $\ge 90\%$**	HiGHS MILP with continuous SoC slack ($z_v \in [0, 1]$, 10,000 € penalty) and 10^{-3} dimensional scaling
Scenario 5: S5 New Bus Depot	Commissioning misconfiguration: `site_config.csv` capped grid limit to 600 kW vs 900 kW physical	5 missed bus departures (`508.95 kWh` uncharged deficit)	**0 missed departures (100% deficit recovered)**	Three-Way Config Reconciler auto-detects discrepancy, restores 900 kW rating and awakens D5-8 eSIM
Scenario 6: S1 Hardware Expansion	Evaluating 12 $\to$ 24 dispenser expansion under 1,080 kW cabinet limit	Doubling plugs alone yields **8 missed departures** (worse!) due to power starvation	**0 missed departures** achieved via cross-cabinet busbar tie	Proves hardware expansion requires cross-cabinet dynamic power routing rather than redundant plugs

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💼 SLA Commercial Model & Economics

1. 20.76 FTE Labor Deconstruction

Analysis of all 8,852 intervention rows in `operations_log.csv` reveals total human labor of 93,412 minutes (1,556.87 hours / 2 weeks), equivalent to **20.76 FTE** (Finnish statutory 37.5 h/wk) costing **1,821,534 €/year** (@ 45.00 €/h):

```
Total Labor Baseline: 20.76 FTE (1,821,534 €/year)
├── [CLAIMED] Software-Addressable: 12.44 FTE (59.95% / 1,091,850 €/year)
│   ├── Midnight Yard Replugging: 9.95 FTE (5,596 events, 47.93% labor time)
│   ├── Manual Grid Load Adjustments: 2.28 FTE (684 events, 10.98% labor time)
│   ├── Remote System Resets: 0.16 FTE (73 events)
│   └── Manual Price Updates: 0.05 FTE (8 events)
├── [RETAINED] Field Technician Dispatches: 5.09 FTE (24.52% / 446,667 €/year)
│   └── 73 physical technician rollouts (cleared_by=technician, component swaps)
└── [RETAINED] Customer Service Contact: 3.22 FTE (15.53% / 282,558 €/year)
    └── 2,418 inbound user inquiries, addressed via self-service kiosk app
```

2. The One-Page Service Level Agreement (One-Page SLA)

Project Sentinel delivers an outcome-based commercial contract backed by mathematical verification: - **Titanium Tier**: $\ge 99.90\%$ Service Level Objective (SLO), $< 50\text{ ms}$ switching reaction, 0-touch autonomous replug. - **Gold Tier**: $\ge 99.50\%$ SLO, $< 100\text{ ms}$ reaction. - **Silver Tier**: $\ge 99.00\%$ SLO. - **Minimum Required Dwell Time (MRDT §5.1) Anti-Moral-Hazard Gate**: Protects operators against fraudulent claims: $$\text{MRDT}*j* = *\left(\frac{\frac{\text{SoC}}*{\text{req}} - \text{SoC}}*{\text{arrive}}}{100}* *\cdot E*{\text{batt}}}{\min(P_{\text{charger}}, P_{\text{veh}}) \cdot 0.95} \right) \cdot 60 + 15.0 \text{ min}$$ - **150% Monthly Liability Cap**: Operators are protected by an absolute monthly payout ceiling ($\le 1.50 \times \text{Monthly Base Fee}$) and guaranteed non-negative monthly billing. - **2029 Customer Story**: Features Pekka Virtanen, Fleet Operations Director at Lahti City Transit (LSL), operating 24 electric articulated buses through sub-zero blizzards with zero depot staff.

⚡ Quick Start & Reproduction Commands

1. Environment Setup

```
# Recommended: Python 3.10+
pip install -r requirements.txt
```

2. Run the Deterministic Verifier (< 0.5s timed execution)

Executes 29 forensic validation routines across all 45 unedited official CSV files and 134,400 power rows:

```
python verify_all.py
```

(Produces deterministic 100% PASS across S1–S5 scenarios, grid invariants, and ROI models in 0.38s of timed computation).

3. Run the S1 Overnight Bus Depot Simulation

Compares ground truth baseline vs. autonomous counterfactual scheduling and cross-cabinet busbar redundancy:

```
python run_s1.py
```

4. Run the Independent Opaque-Box E2E Test Suite (106 Tests)

Validates contract coverage, boundary conditions, cross-feature stress, and Scenario Day workloads:

```
python -m e2e_tests.test_runner
```

(106/106 passed in 0.73s).

5. Run the Comprehensive Pytest Suite (462 Tests)

Executes all unit, integration, protocol, and adversarial simulation tests:

```
pytest tests/ -q
```

(462 passed, 0 failures across 24 test suites).

6. Launch the Interactive 4-View Streamlit Command Center

```
streamlit run app.py
```

- **View 1: Executive Cockpit**: 2029 Customer Story, 20.76 FTE Waterfall & Portfolio SLA.
- **View 2: Scenario Day Live Player**: 00:00–23:45 Scrubber, Live Incidents & Dynamic DPR Matrix.
- **View 3: Hardware Matrix & Protocol Inspector**: 25kW Module Matrix, 260ms BBM Relays & OCPP/ISO Logs.
- **View 4: Commercial SLA & Dispute Clearing**: Interactive MRDT Calculator & Monthly Invoice Export.

7. Launch the 6-Fault Micro-Replay Center

```
streamlit run demo_scenarios/app.py
```

📂 File Tree & Component Functional Dictionary

```
hackson/
├── app.py                                  # Master 4-View Interactive Streamlit Command Cockpit
├── verify_all.py                           # Deterministic verifier running 29 Issue #2 checks in < 0.5s
├── run_s1.py                               # S1 Bus Depot overnight simulation & CAPEX/OPEX accounting
├── requirements.txt                        # Python packaging dependencies
├── pytest.ini                              # Pytest test runner configuration
├── PROJECT.md                              # Single source of engineering truth & technical architecture
├── HACKATHON.md                            # Challenge alignment, competition boundaries & judging rules
├── ZERO_HUMAN_DESIGN.md                    # Zero-human operational design specification
├── One_Page_SLA_Specification.md           # Long-form outcome-based CaaS SLA specification & 2029 story
├── Project_Sentinel_12pages_final.pptx     # 12-page championship presentation pitch deck
├── PITCH_DECK_OUTLINE_v2.md                # Comprehensive pitch deck structure & jury defense guide
├── SUBMISSION_MANIFEST.md                  # SHA-256 cryptographic verification manifest
│
├── kempower-challenge/                     # STRICTLY READ-ONLY OFFICIAL CHALLENGE DATASET
│   └── kempower-challenge-data/            # 45 CSV files, 240,635 rows (unmodified, hash-verified)
│
├── src/                                    # Production core engine & models
│   ├── edf_packer.py                       # Continuous Earliest Deadline First (EDF) discrete simulator
│   ├── core/
│   │   ├── dpr_matrix.py                   # Kempower 25kW DPR dynamic routing matrix & contactor FSM
│   │   ├── mpc_optimizer.py                # 15-minute rolling horizon HiGHS MILP optimizer with slack
│   │   ├── self_healing.py                 # 199-min thermal PdM, 260ms BBM sequencer & edge islanding
│   │   ├── virtual_replug.py               # S1 Strategy A (0 € software) & Strategy B (324k € hardware)
│   │   ├── protocols.py                    # OCPP 2.0.1 Smart Charging & ISO 15118 PnC protocol stack
│   │   └── anti_fraud.py                   # Anti-phantom replug & MRDT §5.1 electrical fingerprint gate
│   ├── sim/
│   │   ├── data_loader.py                  # Official 45 CSV data loader & SHA-256 integrity verifier
│   │   ├── scenario_engine.py              # Closed-loop master scenario execution engine
│   │   ├── discrete_sim.py                 # High-fidelity minute-by-minute discrete event simulator
│   │   ├── night_replay.py                 # S1 overnight charging replay engine
│   │   ├── outage_replay.py                # S3 highway hub WAN outage islanding replay engine
│   │   └── fault_replays.py                # Multi-station fault injection & recovery harnesses
│   ├── commercial/
│   │   ├── roi_analyzer.py                 # 20.76 FTE labor accounting & CAPEX/OPEX payback models
│   │   ├── sla_calculator.py               # Tiered SLA deduction & MRDT anti-moral-hazard engine
│   │   ├── settlement_clearing.py          # B2B monthly billing settlement & invoice clearing
│   │   └── sensitivity_analyzer.py         # Multi-axis NPV/IRR financial sensitivity engine
│   ├── ui/
│   │   └── replay_scene.py                 # Streamlit interactive visualization & rendering components
│   └── utils/
│       └── paths.py                        # Cross-platform deterministic dataset path resolver
│
├── docs/                                   # System whitepapers, reports & audit specifications
│   ├── PROJECT_AUDIT_REPORT.md             # Forensic system audit report across 5 evaluation dimensions
│   ├── energy_impact.md                    # Station-by-station energy ledger & 95-site network proof
│   ├── attribution_matrix.md               # 16-cell S1/S2/S5 causal attribution matrix
│   ├── assumptions_and_additions.md        # Comprehensive dictionary of 20 engineering assumptions
│   ├── one_page_agreement.html             # Formatted One-Page SLA agreement (strictly 1 A4 page)
│   ├── one_page_agreement.md               # Markdown source of One-Page SLA agreement
│   ├── demo_day/                           # Complete Demo Day operational & stage defense manual
│   │   ├── README.md                       # Demo Day master handbook index
│   │   ├── 00_pocket_cheat_sheet.md        # On-stage pocket reference card & golden quotes
│   │   ├── 01_stage_schedule_and_setup.md  # Minute-by-minute stage timing & hardware layout
│   │   ├── 02_team_roles_and_coordination.md # 4-person role matrix, handoff cues & console SOP
│   │   ├── 03_judge_personas_and_question_bank.md # 3 juror profiles & 14 hardcore questions
│   │   ├── 04_structured_defense_and_crisis_manual.md # 3-part structured defense & crisis protocols
│   │   ├── 05_live_demo_operational_sop.md # Live console operator step-by-step SOP
│   │   └── 06_code_proof_and_authenticity_guide.md # Code-level proof & data authenticity guide
│   └── hackathon/                          # Official hackathon requirements, rules & submission guides
│       ├── BRIEF.md                        # Official challenge brief transcription
│       ├── BUILD_PLAN.md                   # Engineering build & delivery plan
```

```
│       ├── DECISIONS.md                    # Architectural decision log
│       ├── DEMO_SCRIPT.md                  # 120-second live presentation script
│       ├── JUDGING.md                      # Official evaluation criteria & weights
│       ├── RULES.md                        # Competition rules & perspective declaration
│       └── SUBMISSION.md                   # Deliverables checklist & submission criteria
│
├── demo_scenarios/                         # Battle-tested failure scenarios & micro-replay videos
│   ├── 1.Night replay.mp4                  # S1 overnight charging & thermal trip replay (1080p)
│   ├── 2.Outage replay.mp4                 # S3 cellular WAN outage & edge islanding replay (1080p)
│   ├── 3.Price spike replay.mp4            # S4 spot price spike (799.30 €/MWh) peak shaving replay
│   ├── 4.Curtailment replay.mp4            # S2 DSO curtailment (400 kW) preemption replay (1080p)
│   ├── 5.Cable fault replay.mp4            # S1 dispenser cable fault & BBM rerouting replay (1080p)
│   ├── 6.Commissioning replay.mp4          # S5 configuration reconciliation replay (1080p)
│   ├── presentation 1.mp4                  # Stage presentation rehearsal video part 1
│   ├── presentation 2.mp4                  # Stage presentation rehearsal video part 2
│   ├── app.py                              # Dedicated 6-scenario micro-replay Streamlit app
│   ├── SPEECH_SIX_SCENES.md                # Comprehensive verbal presentation script for all 6 scenes
│   └── SPEECH_OVERHEAT_SCENE.md            # Deep-dive speech script for S1 thermal trip
│
├── tests/                                  # 462 Pytest automated unit, integration & adversarial tests
│   ├── test_commercial_settlement.py       # SLA calculations, MRDT & monthly clearing tests
│   ├── test_protocols_ocpp_iso.py          # OCPP 2.0.1 smart charging & ISO 15118 PnC tests
│   ├── test_anti_fraud_replug.py           # Anti-phantom replug & electrical fingerprint tests
│   ├── test_discrete_sim_engine.py         # Discrete-event simulation validation tests
│   ├── test_github_issue2_29items.py       # 29-item GitHub Issue #2 audit verification tests
│   ├── test_demo_day_simulation.py         # Demo Day operational manual & role matrix tests
│   ├── test_remediated_sla.py              # SLA piecewise deduction & MRDT tests
│   ├── test_m1_adversarial_probe.py        # Boundary conditions & electrical safety tests
│   ├── test_mpc_math_fixes.py              # HiGHS MILP slack & convex convergence tests
│   └── ...                                 # Additional integration & stress testing modules
│
├── e2e_tests/                              # Independent opaque-box end-to-end testing suite
│   ├── test_runner.py                      # Standalone E2E runner executing all 4 tiers
│   ├── tier1_feature_coverage.py           # Tier 1: Core feature contract tests (45 tests)
│   ├── tier2_boundary_corner.py            # Tier 2: Boundary, stress & corner cases (40 tests)
│   ├── tier3_cross_feature.py              # Tier 3: Pairwise cross-feature interactions (12 tests)
│   └── tier4_real_world_scenario.py        # Tier 4: Scenario Day real-world workloads (9 tests)
│
└── tools/                                  # Engineering analysis & auditing scripts
    ├── cjk_audit.py                        # Automated CJK character scanner guaranteeing zero CJK
    ├── s1_multicabinet_sim.py              # S1 multi-cabinet discrete simulation script
    ├── export_deck_ledger.py               # Presentation deck data ledger generator
    └── check_demo_artifacts.py             # Video & presentation asset validator
```

👥 Competition Team & Acknowledgments

- **Event**: LahtiES Energy Tech Hackathon (LUT University Lahti / CrazyTown, Nov 5–7, 2026)
- **Challenge Sponsor**: Kempower Oyj (Lahti, Finland)
- **Mentors & Jury**: Mats Mattsson, Panu Vartiainen, Iiro Kinnunen, Thet Zin, Chuong Pham, Paul Carlson
- **Integrity Statement**: All 45 official CSV files under `kempower-challenge-data/` remain strictly read-only and hash-verified. All hardware pricing is grounded strictly in `equipment_costs.csv`.